

Question Paper

Exam Date & Time: 23-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 25/JAN 26
SUBJECT: ECE 3121/ECE_3121 ANALOG AND DIGITAL COMMUNICATION
(SPL: ELECTRONICS & COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

Missing data may be suitably assumed.

- 1A) Describe with neat diagrams the scheme for generation and demodulation of VSB modulated wave. (5)
Derive the spectrum of the demodulated signal $v_0(t)$ highlighting the condition of the transfer function $H(f)$. Draw the spectrum of product modulator output $v(t)$ in the demodulation scheme and the spectrum of demodulated signal $v_0(t)$.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) Derive the expression of a wideband FM signal starting from the instantaneous frequency relation, (3)
and show how Bessel functions describe its spectrum.
(i) A 5 kHz audio signal with frequency deviation of 75 kHz is transmitted using FM. Using Carson's Rule, calculate the approximate bandwidth required.
(ii) Recalculate the bandwidth if the frequency deviation is increased to 100 kHz. Compare and comment on the effect of frequency deviation on bandwidth.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Determine the Fourier Transform of $x(t) = \sin(2\pi f_0 t)$. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Consider the signals in the interval $[0, T]$: (5)

$$s_1(t) = \begin{cases} 3, & 0 \leq t \leq T \\ 0, & \text{elsewhere} \end{cases}$$

$$s_2(t) = \begin{cases} 6, & 0 \leq t \leq T/3 \\ 0, & \text{elsewhere} \end{cases}$$

$$s_3(t) = \begin{cases} -3, & 0 \leq t \leq T \\ 0, & \text{elsewhere} \end{cases}$$

Find orthonormal basis functions, and express each signal in terms of them.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) (3)

From the properties of matched filters, apply the relation between the matched filter output and the autocorrelation function (ACF) of the input signal to demonstrate that at $t = T$, the output equals the signal energy E .

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2C) Consider the signal coordinates: (2)

$$s_1 = (1,1), s_2 = (-1,1), s_3 = (-1,-1), s_4 = (1,-1).$$

Draw the constellation diagram and express them as linear combinations of orthonormal basis functions $\Phi_1(t)$ (assume in x - axis) and $\Phi_2(t)$ (assume in y - axis). Extend to Z axis if third basis function is applicable.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3A) Consider the binary data sequence **1 0 0 1 1 0 1** which is applied to the input of a duo binary system. (5)

(i) Determine the output of the duo-binary coder and the corresponding receiver output, without a precoder.

(ii) Determine the output of the duo-binary coder and the corresponding receiver output, with a precoder.

Assume reference bit to be 1.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) Derive the autocorrelation function of the bipolar NRZ code format. Assume statistically independent and equally likely message bits. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3C) With a neat diagram explain the working principle of Delta modulation (DM transmitter) system. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 4A) Derive the expression for Nyquist criterion for distortionless baseband transmission. Also discuss the ideal solution for the same, with the help of relevant mathematical representations and figures. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4B) Derive the expression for the average probability of symbol error for coherent binary PSK detection with all intermediate steps. Assume AWGN channel. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) The binary data sequence **0 1 1 1 1 0 0 1** is applied to the input of a DPSK transmitter. Determine the differentially encoded sequence and transmitted phase. **Assume initial reference bit as 1** (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Construct minimum variance Huffman code for the given source probabilities $p_k = [0.26, 0.34, 0.1, 0.13, 0.17]$. Also determine the average code length of this Huffman code. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) Determine the entropy, efficiency and redundancy of the Huffman code as given in Q5A. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

5C) Determine the minimum variance of this Huffman code as given in Q5A. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		ECE_3121:ANALOG AND DIGITAL COMMUNICATION	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		ECE_3121:ANALOG AND DIGITAL COMMUNICATION	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	10	35(70%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)